Alexander Arens

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EDUCATION

Purdue University — **West Lafayette, IN**
Bachelor of Science, Electrical and Computer Engineering — **December 2027**

PROFESSIONAL EXPERIENCE

Fuse Medical — **Dallas, TX**
Interim Engineer — **May 2024 – Aug 2026**

- Assumed lead engineering responsibilities, applying DFM to replace a machined assembly with a novel linkage, cutting costs 30x.
- Engineered parametric models for legacy surgical tools, improving ergonomic design in direct response to surgeon feedback.
- Redesigned SolidWorks CAD templates to improve efficiency and standardization.

PERSONAL PROJECTS

- Full portfolio link https://flukeychip.github.io/Project-Portfolio/
- **Autonomous Drone Defense (Purdue AI Hackathon):** Led a team building an autonomous targeting system that detects, classifies, and tracks airborne threats, with a classifier that inhibits the laser near living things.
- **Scholarship Recipient – All-In Summit:** Selected as a scholarship recipient for the All-In Summit in California from a pool of thousands of applicants. Engaged with leaders in technology, medical, business, and government sectors. Invited to contribute to a medical AI startup.
- **6DOF Arm:** Inspired by the Universal Robots UR3e arms in a campus design lab, designed and built my own six-axis arm with 3D printed 20:1 cycloidal drives, carbon fiber struts, belt driven remote actuation, magnetic Hall effect encoders, and an NVIDIA Jetson.
- **NFC PCB Business Card:** Designed a PCB business card with an integrated NFC antenna. Harvests power from a phones transmitter, and then streams a landing page back to the phone,
- **Rapidly deployable quadcopter:** Led a team to develop a drone with a compact, easily manufacturable design, a hot-swappable camera system, and ultrasonic sensor-based obstacle avoidance.
- **Smart Drink Dispenser Table:** Developed a device designed for discreet integration into a tabletop. Engineered the system to autonomously prepare beverages in response to user commands.
- **Automated home security:** Led a team of three engineers to design a pan-tilt camera system with IR illuminators, lasers, and cameras that detect and track people through a full 360° field. Programmed image recognition with closed-loop visual feedback, including PD control, adaptive dead zones, and platform-specific optimizations.
- **Monodirectional speaker:** Designed a circuit that amplitude-modulates a 40 kHz ultrasonic carrier with audible content.
- **Active orthosis (ASME):** Collaborated with the electronics division of an active orthopedic device that allowed patients with nerve or muscular damage to arms to restore movements using a multitude of sensors and actuators

RELEVANT SKILLS

- **Electronics:** PCB design and layout (KiCad, Autodesk Eagle), microcontrollers (RP2040, ESP32, ATmega), analog and digital circuit analysis
- **Software & Programming:** C, C++, Python, computer vision (OpenCV), FPGA (Verilog, iCE40), MATLAB
- **Mechanical & CAD:** Fusion 360, SolidWorks, NX
- **Prototyping:** Additive manufacturing (FDM, SLA, SLS), SMT assembly (solder paste, reflow)

LEADERSHIP AND INVOLVEMENT

Eagle Scout | Phi Delta Theta Project Manager, Assistant House Manager | 3D Printing Club | Purdue ASME